Inorganic CUME

Jan-24-2014

1. Which element's reduction half reaction gives the most positive *emf*, and which gives the most negative *emf*? (*10 points*)

2. Identify the element(s) having the highest melting point in each row of the transition metals. (*15 points*)

3. What would be the alkali metals' general trend(s) of chemical properties? (*10 points*)

4. Rank each group of the molecules in the order of increasing acidity (*10 points*)
   i) SiH_4, CH_4, GeH_4
   ii) Al_2O_3, MgO, Na_2O, SiO_2

5. Define two reactions, each for making a halogen gas in lab. (*10 points*)

6. Find one Zn chalcogenide for each chalcogen, and predict which chalcogenide would be semiconducting. (*15 points*)

7. Chemistry and Structures of Carbon (*20 points*):
   (i). Indicate which carbon structure(s) would be conductive, and/or semiconducting, and/or insulating.
   (ii) Explain why some would be colored (or colorless).